

# Anjila Belbase

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## Education

### University of South Florida

Bachelor of Science in Electrical Engineering

FE Exam Scheduled – Jan. 9, 2026

Tampa, FL

December 2025

## Technical Skills

**Automation & Controls:** Allen-Bradley PLCs, Studio 5000, RSLogix 500, Ladder Logic, FactoryTalk View, HMIs, Industrial I/O

**Programming & Embedded:** Python, C/C++, MATLAB, JavaScript, ESP32, MQTT, REST APIs

**Design, Tools & Testing:** AutoCAD Electrical, Fusion 360, BOMs, SOPs, React, Git, VS Code, Microsoft Word, Excel, PowerPoint, Digital Multimeter, Clamp Meter, Voltage Tester, Oscilloscope

## Experience

### Controls Engineering Specialist

August 2024 – December 2025

*Cirkul Inc.*

*Tampa, FL*

- Orchestrated end-to-end controls integration for an industrial water treatment line, delivering 6+ electrical/controls artifacts and interfacing a vendor RO unit with mineral injection, ozone treatment, and 4+ pressure/pH sensors.
- Standardized BOMs, electrical schematics, and SOPs in AutoCAD, cutting rework and documentation errors by 25%.
- Architected a real-time PLC status and alarm system, reducing troubleshooting delays by 15%.
- Re-engineered and commissioned dryer and washer PLC systems, correcting I/O and safety interlocks to reduce cycle variability by 10%.
- Resolved chronic equipment failures through root-cause analysis and targeted PLC logic updates, increasing reliability by 15% through cross-functional collaboration.
- Accelerated troubleshooting and inspection by 25% using Studio 5000, RSLogix 500, Productivity Suite, FactoryTalk View, and C-More HMI, with Cognex vision integration.

### Electrical Engineering Intern

May 2024 – August 2024

*Cirkul Inc.*

*Tampa, FL*

- Executed hands on field wiring for control panels and production equipment across multiple manufacturing lines, supporting 3+ controls upgrade projects alongside industrial electricians.
- Generated and revised AutoCAD electrical schematics and control panel layout drawings for 20+ panels and system updates, improving documentation accuracy and build readiness.
- Supported electrical panel assembly, wire labeling, and I/O point verification for 100+ connections, helping reduce installation errors and rework during commissioning.

## Projects

### Smart IoT Farming System for Microgreens

January 2025 – December 2025

*Tools: ESP32, C++, MQTT, Wi-Fi, NuvIoT Cloud, Fusion 360 (3D Printing)*

- Co-developed an ESP32 IoT system with a 3-member team, streaming temperature, humidity, CO<sub>2</sub>, soil moisture, pressure, and light data via MQTT every 30s, achieving >98% uptime.
- Drove firmware, MQTT tuning, and enclosure design to deliver >98% transmission reliability, sub-second cloud latency, and reliable operation in humid conditions.

### JamGesture: IMU-Based Gesture-Controlled MP3 System

March 2025 – May 2025

*Tools: ESP32, IMU, C, Kalman Filter, DFPlayer Mini, Fusion 360 (3D Printing)*

- Engineered ESP32 IMU gesture control for MP3 playback and volume, achieving >95% recognition accuracy.
- Stabilized motion sensing via Kalman filtering and enclosure optimization, improving IMU stability 60% and reducing motion noise 35%.

## Certificates & Achievements

Engineer-in-Training (EIT) — Pending (FE Exam: Jan 9, 2026)